1 Updated equations (bundled form)

The equations that were updated to work with available data.

1. Loss of cytoplasmic Smoothened equation

$$removed \rightarrow \sum \text{influx} = k_{in} \cdot SMO_{cill} \tag{1}$$

   Constant influx term as opposed to true exchange term. Substitution to remove unknown parameters and readapt to known constants.

2. Michaelis-Menten MMM to linear mass-action

$$\begin{aligned} \frac{dSMO_{cil}}{dt} = \sum \text{influx} - k_{out}SMO_{cil} - k_{deg}SMO_{cil} - k_{MMM} \cdot Mm_{free} \cdot SMO_{cil} \\ + \alpha_L \cdot k_{MMM} \cdot Mm_{link} - k_{PTCH1} \cdot PTCH1 \cdot SMO_{cil} \end{aligned} \tag{2}$$

   Because MGRN1 is an E3 ligase accuracy of it as a single enzymatic agent (as is expected in Michaelis-Menten) is uncertain. MMM lumped term is replaced with linear mass action kinetic relationships.

   This also includes the addition of a ternary term for PROTAC behavior. This term (α_L), is the accumulation of the Linker. Because it is untested we will sweep over multiple conditions to see tentative functionalities at varying rates.

3. Mass-action form of Ubiquitinated SMO

$$\frac{dSMO_{ub}}{dt} = k_{MMM}M_{free}SMO_{cil} + \alpha_L k_{MMM}M_{link}SMO_{cil} \tag{3}$$

   Updated PROTAC arm with simplified ternary constant form.